

Reconstruction of the Collatz-type Map ($3n + 5$ Problem) via Jacobsthal Normalized Spinor Representation

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Abstract

In this study, we propose a novel number-theoretic framework for the generalized Collatz map, where the odd nuclei are normalized by the Jacobsthal (J) sequence and expressed as two-component discrete spinors (J-spinors) $J_c(a, b)$.

$$T(n) = \begin{cases} n/2 & (n \text{ even}) \\ 3n + 5 & (n \text{ odd}) \end{cases}$$

The core of this research lies in redefining the transition of odd nuclei not as the conventional procedural operation of " $3n + 5 \rightarrow \text{divide by } 2^c$ ", but as a linear transition of J-spinors. An arbitrary odd nucleus N_{current} is uniquely expressed as follows:

$$N_{\text{current}} = J_c(a, b)$$

Furthermore, for the next odd nucleus N_{next} , the following equation strictly holds:

$$N_{\text{next}} = a + b = \frac{3J_c + 5}{2^c}$$

Consequently, it has been revealed that the complex odd-nucleus dynamics of the $3n + 5$ map are perfectly embedded within the linear structure of the J-sequence.

1. J-spinor Expression

By expanding the odd nucleus N_{current} into the Jacobsthal (J) sequence, it can be uniquely represented as a two-component discrete spinor (J-spinor) $J_c(a, b)$. Here, (a, b) are defined as the J-wave source, and c as the J-index (spatial shift count).

1.1 J-sequence

For any wave source (a, b) , the J-sequence J_m is generated by the following recurrence relation:

$$J_m = J_{m-1} + 2J_{m-2}, \quad (J_0, J_1) = (a, b)$$

1.2 Reconstruction of J-spinor

Assuming the odd nucleus N_{current} to be the term J_c of the J-sequence, the corresponding wave source (a, b) is uniquely reconstructed through the following procedure.

- 1) Setting the reference term:

$$J_c = N_{\text{current}}$$

- 2) Defining the subordinate term specific to the $3n + 5$ system:

$$J_{c-1} = \frac{N_{\text{current}} + 5}{2}$$

- 3) Reconstructing subordinate terms via the inverse recurrence relation:

$$J_{m-2} = \frac{J_m - J_{m-1}}{2}$$

- 4) Setting the wave source (minimum adjacent terms):

$$(a, b) = (J_0, J_1)$$

Here, J_0 is the lowest term as an integer.

- 5) J-spinor representation:

$$N_{\text{current}} = J_c(a, b)$$

1.3 Next Odd Nucleus

The wave source (a, b) directly yields the next odd nucleus N_{next} :

$$N_{\text{next}} = a + b$$

Due to the structure of the J-sequence, the sum of adjacent terms always grows by a scale factor of 2:

$$J_m + J_{m-1} = 2(J_{m-1} + J_{m-2})$$

1.4 Equivalence with the $3n + 5$ Map

The J-sequence expands as follows:

$$a, b, (2a + b), (2a + 3b), (6a + 5b), \dots$$

The sum of its adjacent terms is always expressed as:

$$(a + b) \cdot 2^m$$

Meanwhile, the recurrence relation for the odd nucleus in the $3n + 5$ map is:

$$3N_{\text{current}} + 5 = N_{\text{next}} \cdot 2^c$$

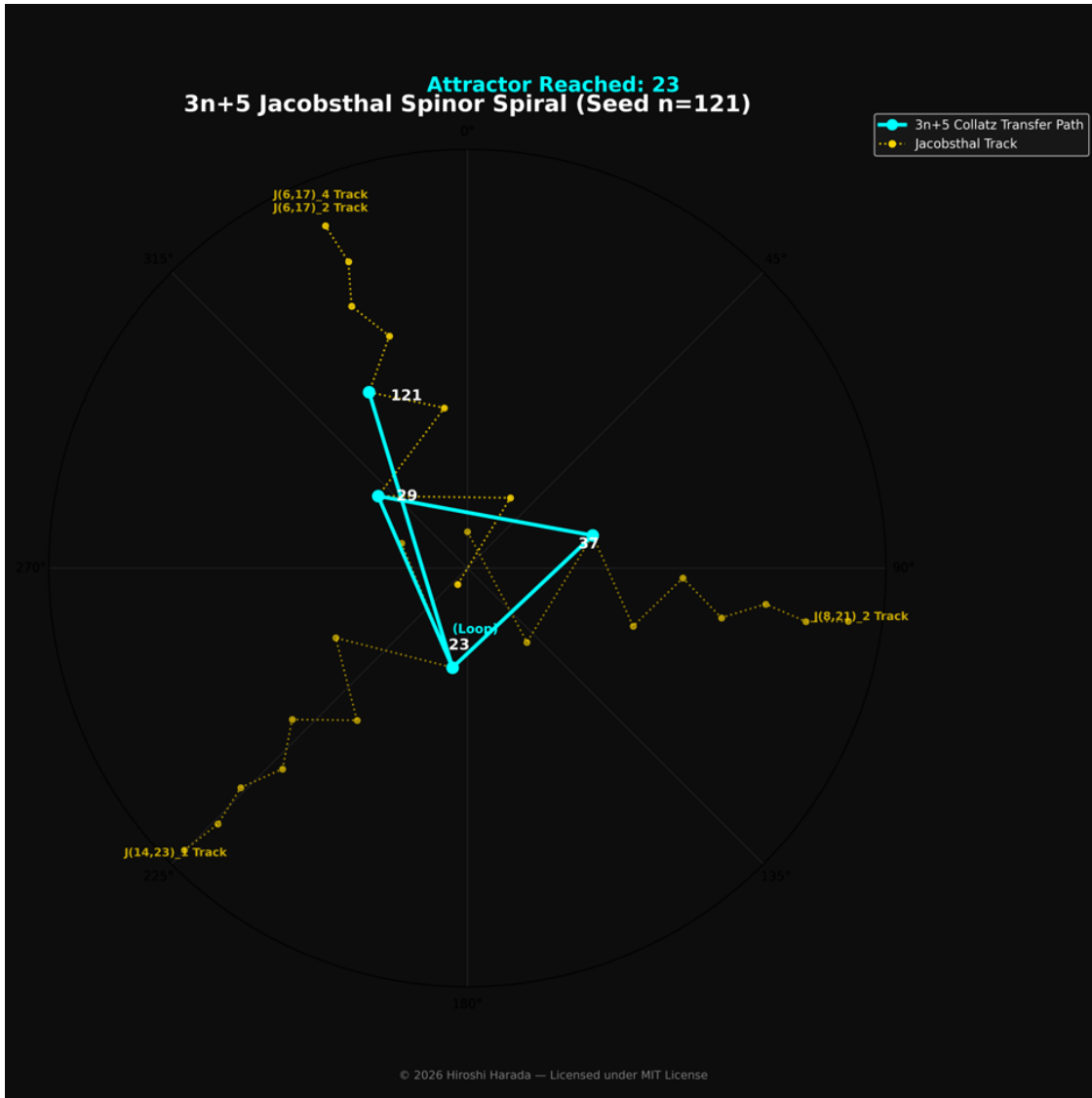
By definition, the subordinate term J_{c-1} of N_{current} is:

$$J_{c-1} = \frac{N_{\text{current}} + 5}{2}$$

From the above, the following always holds:

$$3N_{\text{current}} + 5 = 2(J_c + J_{c-1}) = (a + b) \cdot 2^c = N_{\text{next}} \cdot 2^c$$

Thus, it has been demonstrated that the odd-nucleus transition of the $3n + 5$ map is completely encapsulated within the linear structure of the J-spinor. **Figure 1** illustrates an example of the odd Collatz trajectory with an initial value of 121.



Odd	$J_m(a,b)$	$a+b$	$m = 0$	1	2	3	4	5	6	7	8
121	$J_4(6,17)$	23	6	17	29	63	121	247	489	983	1961
23	$J_1(14,23)$	37	14	23	51	97	199	393	791	1577	3159
37	$J_2(8,21)$	29	8	21	37	79	153	311	617	1239	2473
29	$J_2(6,17)$	23	6	17	29	63	121	247	489	983	1961

Figure 1. Visualization of the J-spinor Trajectory in Logarithmic Spiral Space.

Logarithmic spiral mapping of the odd-nucleus trajectory (cyan) and Jacobsthal (J) tracks (gold) for an initial value of $n = 121$. The visualization demonstrates the geometric topology of the sequence entering the attractor "23" from an external portal (121) via a deep spatial shift ($c = 4$) and subsequently closing as a localized standing wave.

2. Verification: Six Attractor Systems

The $3n + 5$ map possesses the following six distinct odd-nucleus attractors:

$$1, 5, 19, 23, 187, 347$$

For each attractor, we verify the topological closure of the wave source via the J-spinor.

2.1 Attractor 1

The odd-nucleus trajectory with an initial value of 9 converges to attractor 1.

$$9 \rightarrow 1 \rightarrow 1 \text{ (Loop)}$$

2.1.1 Hop 1: $9 \rightarrow 1$

- $3 \cdot 9 + 5 = 32 \Rightarrow 32/2^5 = 1 \text{ } (c = 5)$
- $J_c = 9, J_{c-1} = (9 + 5)/2 = 7, J_{c-2} = (9 - 7)/2 = 1, J_{c-3} = (7 - 1)/2 = 3, J_{c-4} = (1 - 3)/2 = -1, J_{c-5} = (3 - (-1))/2 = 2$
- $J(2, -1): 2, -1, 3, 1, 7, 9, 23, \dots$
- $J_5(2, -1) = 9$
- $N_{\text{next}} = 2 + (-1) = 1$

2.1.2 Hop 2: $1 \rightarrow 1$

- $3 \cdot 1 + 5 = 8 \Rightarrow 8/2^3 = 1 \text{ } (c = 3)$
- $J_c = 1, J_{c-1} = (1 + 5)/2 = 3, J_{c-2} = (1 - 3)/2 = -1, J_{c-3} = (3 - (-1))/2 = 2$
- $J(2, -1): 2, -1, 3, 1, 7, 9, 23, \dots$
- $J_3(2, -1) = 1$
- $N_{\text{next}} = 2 + (-1) = 1$

In Hop 2, the wave source strictly returns to $J(2, -1)$, completely identical to Hop 1, establishing a perfect topological closure in the J-spinor space.

2.2 Attractor 5

The odd-nucleus trajectory with an initial value of 25 converges to attractor 5.

$$25 \rightarrow 5 \rightarrow 5 \text{ (Loop)}$$

2.2.1 Hop 1: 25 → 5

- $3 \cdot 25 + 5 = 80 \Rightarrow 80/2^4 = 5 \text{ (} c = 4 \text{)}$
- $J_c = 25, J_{c-1} = (25 + 5)/2 = 15, J_{c-2} = (25 - 15)/2 = 5, J_{c-3} = (15 - 5)/2 = 5, J_{c-4} = (5 - 5)/2 = 0$
- $J(0,5) = 5J(0,1): 0, 5, 5, 15, 25, 55, \dots$
- $J_4(0,5) = 25$
- $N_{\text{next}} = 0 + 5 = 5$

2.2.2 Hop 2: 5 → 5

- $3 \cdot 5 + 5 = 20 \Rightarrow 20/2^2 = 5 \text{ (} c = 2 \text{)}$
- $J_c = 5, J_{c-1} = (5 + 5)/2 = 5, J_{c-2} = (5 - 5)/2 = 0$
- $J(0,5) = 5J(0,1): 0, 5, 5, 15, 25, 55, \dots$
- $J_2(0,5) = 5$
- $N_{\text{next}} = 0 + 5 = 5$

In Hop 2, the wave source perfectly returns to $J(0,5)$, which is the entry state of Hop 1.

Interestingly, the J-spinor upon convergence to attractor 5 was $J(0,5)$, which is precisely the fundamental wave $J(0,1)$ of the standard system scaled by a factor of 5.

Conversely, the J-spinor upon convergence to attractor 1 was $J(2, -1)$.

Considering that in the $3n + 1$ system, the J-spinor for the absolute attractor 1 is $J(0,1)$, it implies that "1" in the $3n + 5$ system is nothing more than one of the localized standing wave modes.

Therefore, it is suggested that the true ground state in the $3n + 5$ system is "5".

This structural variance provides crucial topological insight into how altering the additive term in Collatz-type maps displaces the basis of the J-space.

2.3 Attractor 19

The odd-nucleus trajectory with an initial value of 11 converges to attractor 19.

$$11 \rightarrow 19 \rightarrow 31 \rightarrow 49 \rightarrow 19 \text{ (Loop)}$$

2.3.1 Hop 1: 11 → 19

- $3 \cdot 11 + 5 = 38 \Rightarrow 38/2^1 = 19 \text{ (} c = 1 \text{)}$
- $J_c = 11, J_{c-1} = (11 + 5)/2 = 8$
- $J(8,11)$: 8, 11, 27, 49, ...
- $J_1(8,11) = 11$
- $N_{\text{next}} = 8 + 11 = 19$

2.3.2 Hop 2: 19 → 31

- $3 \cdot 19 + 5 = 62 \Rightarrow 62/2^1 = 31 \text{ (} c = 1 \text{)}$
- $J_c = 19, J_{c-1} = (19 + 5)/2 = 12$
- $J(12,19)$: 12, 19, 43, 81, ...
- $J_1(12,19) = 19$
- $N_{\text{next}} = 12 + 19 = 31$

2.3.3 Hop 3: 31 → 49

- $3 \cdot 31 + 5 = 98 \Rightarrow 98/2^1 = 49 \text{ (} c = 1 \text{)}$
- $J_c = 31, J_{c-1} = (31 + 5)/2 = 18$
- $J(18,31)$: 18, 31, 67, 129, ...
- $J_1(18,31) = 31$
- $N_{\text{next}} = 18 + 31 = 49$

2.3.4 Hop 4: 49 → 19

- $3 \cdot 49 + 5 = 152 \Rightarrow 152/2^3 = 19 \text{ (} c = 3 \text{)}$
- $J_c = 49, J_{c-1} = (49 + 5)/2 = 27, J_{c-2} = (49 - 27)/2 = 11, J_{c-3} = (27 - 11)/2 = 8$
- $J(8,11)$: 8, 11, 27, 49, ...
- $J_3(8,11) = 49$
- $N_{\text{next}} = 8 + 11 = 19$

Through a 3-step loop, the wave source closes at $J(8,11)$ (completely identical to the wave source in Hop 1).

2.4 Attractor 23

The odd-nucleus trajectory with an initial value of 121 converges to attractor 23.

$$121 \rightarrow 23 \rightarrow 37 \rightarrow 29 \rightarrow 23 \text{ (Loop)}$$

2.4.1 Hop 1: 121 $\rightarrow$ 23

- $3 \cdot 121 + 5 = 368 \Rightarrow 368/2^4 = 23 \text{ (} c = 4 \text{)}$
- $J_c = 121, J_{c-1} = (121 + 5)/2 = 63, J_{c-2} = (121 - 63)/2 = 29,$
 $J_{c-3} = (63 - 29)/2 = 17, J_{c-4} = (29 - 17)/2 = 6$
- $J(6,17): 6, 17, 29, 63, 121, \dots$
- $J_4(6,17) = 121$
- $N_{\text{next}} = 6 + 17 = 23$

2.4.2 Hop 2: 23 $\rightarrow$ 37

- $3 \cdot 23 + 5 = 74 \Rightarrow 74/2^1 = 37 \text{ (} c = 1 \text{)}$
- $J_c = 23, J_{c-1} = (23 + 5)/2 = 14$
- $J(14,23): 14, 23, 51, 97, \dots$
- $J_1(14,23) = 23$
- $N_{\text{next}} = 14 + 23 = 37$

2.4.3 Hop 3: 37 $\rightarrow$ 29

- $3 \cdot 37 + 5 = 116 \Rightarrow 116/2^2 = 29 \text{ (} c = 2 \text{)}$
- $J_c = 37, J_{c-1} = (37 + 5)/2 = 21, J_{c-2} = (37 - 21)/2 = 8$
- $J(8,21): 8, 21, 37, 79, \dots$
- $J_2(8,21) = 37$
- $N_{\text{next}} = 8 + 21 = 29$

2.4.4 Hop 4: 29 $\rightarrow$ 23

- $3 \cdot 29 + 5 = 92 \Rightarrow 92/2^2 = 23 \text{ (} c = 2 \text{)}$
- $J_c = 29, J_{c-1} = (29 + 5)/2 = 17, J_{c-2} = (29 - 17)/2 = 6$
- $J(6,17): 6, 17, 29, 63, \dots$
- $J_2(6,17) = 29$
- $N_{\text{next}} = 6 + 17 = 23$

Through a 3-step loop, the wave source closes at $J(6,17)$ (completely identical to the wave source in Hop 1).

2.5 Attractor 187

The odd-nucleus trajectory with an initial value of 123 converges to attractor 187.

$$123 \rightarrow 187 \rightarrow 283 \rightarrow 427 \rightarrow 643 \rightarrow 967 \rightarrow 1453 \rightarrow 1091 \rightarrow 1639 \rightarrow 2461 \\ \rightarrow 1847 \rightarrow 2773 \rightarrow 2081 \rightarrow 781 \rightarrow 587 \rightarrow 883 \rightarrow 1327 \rightarrow 1993 \rightarrow 187 \text{ (Loop)}$$

Here, as a representative example, we present the origin of the trajectory and the final Hop where the topology forms a closure.

2.5.1 Hop 1: 123 → 187

- $3 \cdot 123 + 5 = 374 \Rightarrow 374/2^1 = 187 \text{ (} c = 1 \text{)}$
- $J_c = 123, J_{c-1} = 64$
- $J(64,123)$: 64, 123, 251, 497, ...
- $J_1(64,123) = 123$
- $N_{\text{next}} = 64 + 123 = 187$

2.5.2 Hop 2: 187 → 283

- $3 \cdot 187 + 5 = 566 \Rightarrow 566/2^1 = 283 \text{ (} c = 1 \text{)}$
- $J_c = 187, J_{c-1} = (187 + 5)/2 = 96$
- $J(96,187)$: 96, 187, 379, 753, ...
- $J_1(96,187) = 187$
- $N_{\text{next}} = 96 + 187 = 283$

2.5.3 Hop 18: 1993 → 187

- $3 \cdot 1993 + 5 = 5984 \Rightarrow 5984/2^5 = 187 \text{ (} c = 5 \text{)}$
- $J_c = 1993, J_{c-1} = 999, J_{c-2} = 497, J_{c-3} = 251, J_{c-4} = 123, J_{c-5} = 64$
- $J(64,123)$: 64, 123, 251, 497, 999, 1993, ...
- $J_5(64,123) = 1993$
- $N_{\text{next}} = 64 + 123 = 187$

After a vast 17-step loop, the wave source closes at $J(64,123)$, perfectly reverting to the entry state of Hop 1.

2.6 Attractor 347

The odd-nucleus trajectory with an initial value of 1849 converges to attractor 347.

$$1849 \rightarrow 347 \rightarrow 523 \rightarrow 787 \rightarrow 1183 \rightarrow 1777 \rightarrow 667 \rightarrow 1003 \rightarrow 1507 \rightarrow 2263 \\ \rightarrow 3397 \rightarrow 2549 \rightarrow 1913 \rightarrow 359 \rightarrow 541 \rightarrow 407 \rightarrow 613 \rightarrow 461 \rightarrow 347 \text{ (Loop)}$$

2.6.1 Hop 1: 1849 $\rightarrow$ 347

- $3 \cdot 1849 + 5 = 5552 \Rightarrow 5552/2^4 = 347$ ($c = 4$)
- $J_c = 1849$, $J_{c-1} = (1849 + 5)/2 = 927$, $J_{c-2} = (1849 - 927)/2 = 461$,
 $J_{c-3} = (927 - 461)/2 = 233$, $J_{c-4} = (461 - 233)/2 = 114$
- $J(114,233)$: 114, 233, 461, 927, 1849, ...
- $J_4(114,233) = 1849$
- $N_{\text{next}} = 114 + 233 = 347$

2.6.2 Hop 2: 347 $\rightarrow$ 523

- $3 \cdot 347 + 5 = 1046 \Rightarrow 1046/2^1 = 523$ ($c = 1$)
- $J_c = 347$, $J_{c-1} = (347 + 5)/2 = 176$
- $J(176,347)$: 176, 347, 699, 1393, ...
- $J_1(176,347) = 347$
- $N_{\text{next}} = 176 + 347 = 523$

2.6.3 Hop 18: 461 $\rightarrow$ 347

- $3 \cdot 461 + 5 = 1388 \Rightarrow 1388/2^2 = 347$ ($c = 2$)
- $J_c = 461$, $J_{c-1} = (461 + 5)/2 = 233$, $J_{c-2} = (461 - 233)/2 = 114$
- $J(114,233)$: 114, 233, 461, 927, 1849, ...
- $J_2(114,233) = 461$
- $N_{\text{next}} = 114 + 233 = 347$

Similarly, after a vast 17-step loop, the wave source closes at $J(114,233)$, perfectly reverting to the entry state of Hop 1.

3. Conclusion

This study demonstrates that the odd-nucleus transition of the generalized Collatz map $3n + 5$ can be entirely reconstructed as a linear structure of two-component spinors (J-spinors) obtained through J-normalization.

The J-spinor integrates the following elements:

- J-sequence
- Wave source (a, b)
- Shift count c
- Odd nucleus N_{current}

In particular, the following three equations universally hold:

$$N_{\text{current}} = J_c(a, b)$$

$$N_{\text{next}} = a + b$$

$$N_{\text{next}} = \frac{3J_c(a, b) + 5}{2^c}$$

Consequently, the complex odd-nucleus dynamics of the $3n + 5$ map can be comprehended as a perfectly linearized algebraic structure. This method is applicable to any arbitrary odd nucleus and provides a novel number-theoretic representation (spinor-based representation) that eliminates conventional arithmetic operations for the $3n + 5$ problem, which harbors multiple attractors. Furthermore, the empirical results of this paper indicate that the true ground state in the $3n + 5$ system is $J(0,5)$ —which is the fundamental wave $J(0,1)$ of the standard system scaled by a factor of 5—namely "5". In contrast, "1", the former absolute convergence point of the $3n + 1$ system, is no longer a privileged center but merely one of the localized standing wave modes $J(2, -1)$. This structural variance is a vital topological finding that illustrates how modifying the additive term in Collatz-type maps displaces the basis of the Jacobsthal space.

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